MCQ 1: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. A probability distribution has a sum of probabilities equal to
   - A. 0
   - B. 1
   - C. n
   - D. the mean
2. X is 0 or 4 with probabilities $\frac{3}{4}$ and $\frac{1}{4}$; $E(X) =$
   - A. 1
   - B. 2
   - C. 3
   - D. 4
3. $V(X)$ equals
   - A. $E(X^2)$
   - B. $E(X)^2$
   - C. $E(X^2) + E(X)^2$
   - D. $E(X^2) - E(X)^2$
4. If $V(X) = 9$, $\sigma(X) =$
   - A. 81
   - B. 9
   - C. 3
   - D. $\frac{1}{3}$
5. $E(2X + 1)$ equals
   - A. $2E(X)$
   - B. $2E(X) + 1$
   - C. $E(X) + 1$
   - D. $E(X)^2 + 1$

MCQ 2: understand and choose

Five single-answer questions for practice. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official EAM format.

1. $V(X+5)$ equals
   A. $V(X)+5$
   B. $V(X)+25$
   C. $V(X)$
   D. $5V(X)$
2. $\sigma(-3X)$ equals
   A. $3\sigma(X)$
   B. $-3\sigma(X)$
   C. $9\sigma(X)$
   D. $\sigma(X)$
3. The expectation is necessarily
   A. a possible value
   B. an integer
   C. positive
   D. none of these three statements
4. $E(X+Y)=E(X)+E(Y)$ requires
   A. independence
   B. no independence
   C. X=Y
   D. X positive
5. If $E(X)=2$ and $E(X^2)=5$, then $V(X)=$
   A. 3
   B. 9
   C. 1
   D. 7

True or false? Justify.

1. A variance can be negative.
2. Two variables with the same expectation have the same distribution.
3. Adding a constant to a variable does not change its standard deviation.

> **Key point · Self-assessment.** I build a complete distribution; I distinguish gross from net; I check both variance formulas; I do not confuse frequency with probability.
>
> For each objective, indicate: achieved without help; achieved with help; to be reworked. Write one example and one mistake to avoid.